



June 19, 2013

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

○ 12M3C – Mr Lin

[illegible]

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer sheet provided.

Questions

Marks

1. Which of the following exists at the point $(1, 1)$ on the graph of $y = (x - 1)^3 + 1$? 1

- (A) A local minimum. (C) A stationary point of inflexion.
(B) A local maximum. (D) None of the above.

2. The midpoint of the line joining $(0, -5)$ to $(d, 0)$ is 1

- (A) $\left(\frac{d-5}{2}, 0\right)$ (C) $\left(0, \frac{5-d}{2}\right)$
(B) $\left(\frac{d}{2}, -\frac{5}{2}\right)$ (D) $\left(\frac{5+d}{2}, 0\right)$

3. Which of the following is the derivative of 1

$$y = \log_e(f(x))$$

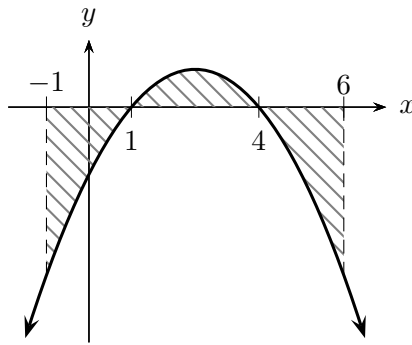
with respect to x ?

- (A) $\frac{f(x)}{f'(x)}$ (C) $\frac{f'(x)}{f(x)}$
(B) $\frac{1}{f'(x)}$ (D) $\frac{1}{f(x)}$

4. What is the period of the function $y = 4 \sin\left(\frac{x}{3}\right)$? 1

- (A) 6π (C) 4
(B) $\frac{2\pi}{3}$ (D) $\frac{1}{4}$

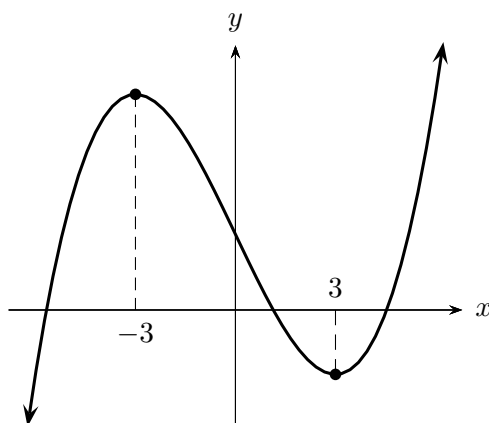
5. The graph with equation $y = x^2$ is translated 3 units down and 2 units to the right. Which equation represents the resulting graph? 1
- (A) $y = (x - 2)^2 + 3$ (C) $y = (x + 2)^2 + 3$
- (B) $y = (x - 2)^2 - 3$ (D) $y = (x + 2)^2 - 3$
6. Which of the following expressions gives the total area of the shaded region in the diagram? 1



- (A) $\int_{-1}^6 f(x) dx$
- (B) $-\int_{-1}^0 f(x) dx + \int_0^6 f(x) dx$
- (C) $-\int_{-1}^1 f(x) dx + \int_1^4 f(x) dx - \int_4^6 f(x) dx$
- (D) $\int_1^4 f(x) dx + 2 \int_4^6 f(x) dx$

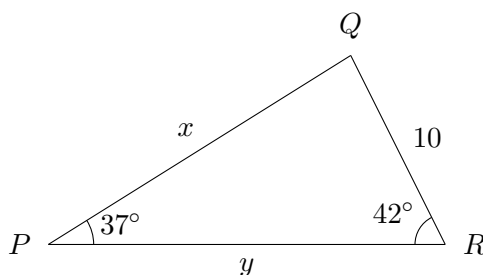
7. From the graph of $y = f(x)$, when is $f'(x)$ negative?

1



- (A) $x < -3$ or $x > 3$ (C) $x \leq -3$ or $x \geq 3$
 (B) $-3 < x < 3$ (D) $-3 \leq x \leq 3$
8. $\triangle PQR$ has side lengths x , y and 10 as shown. $\angle RPQ = 37^\circ$ and $\angle QRP = 42^\circ$.

1



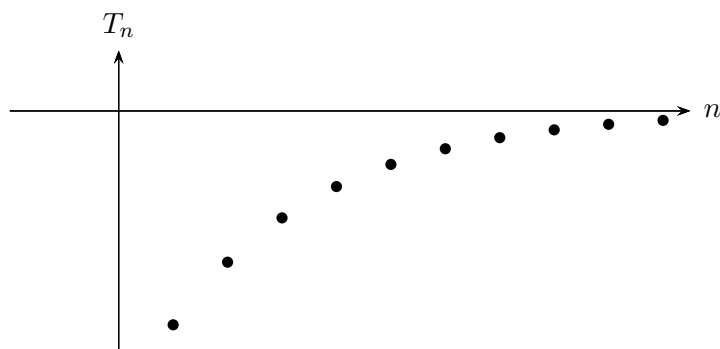
Which of the following expressions is correct for $\triangle PQR$?

- (A) $x = 10 \times \frac{\sin 42^\circ}{\sin 37^\circ}$ (C) $x = \frac{10}{\sin 37^\circ}$
 (B) $y = 10 \times \frac{\sin 37^\circ}{\sin 101^\circ}$ (D) $y = \frac{10}{\tan 37^\circ}$

9. If M is decreasing at an increasing rate, what does this suggest about $\frac{dM}{dt}$ and $\frac{d^2M}{dt^2}$? 1

- (A) $\frac{dM}{dt} < 0$ and $\frac{d^2M}{dt^2} < 0$ (C) $\frac{dM}{dt} < 0$ and $\frac{d^2M}{dt^2} > 0$
- (B) $\frac{dM}{dt} > 0$ and $\frac{d^2M}{dt^2} < 0$ (D) $\frac{dM}{dt} > 0$ and $\frac{d^2M}{dt^2} > 0$

10. The graph shows consecutive terms of a sequence. Which of the following statements best describes the sequence? 1



- (A) geometric, $|r| \geq 1$. (C) arithmetic, $|d| < 1$.
- (B) arithmetic, $|d| \geq 1$. (D) geometric, $|r| < 1$.

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Commence a NEW page.	Marks
(a) Fully factorise $4x^2 - 36$.		2
(b) Solve $ 2x - 3 < 13$.		2
(c) Solve for x : $4^x - 9 \times 2^x + 8 = 0$.		2
(d) For the parabola $(x - 2)^2 = 4y$ i. Find the coordinates of the vertex.		1
ii. State the equation of the directrix of the parabola.		1
(e) Write down the domain of $f(x) = \frac{1}{(x - 3)(2 - x)}$.		2
(f) Evaluate $\lim_{x \rightarrow 2} \frac{x - 2}{x^2 + x - 6}$.		2
(g) Find the equation of the tangent to the curve $y = 2 \sin 2x$ at the point $\left(\frac{\pi}{8}, \sqrt{2}\right)$.		3

Question 12 (15 Marks)

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Marks

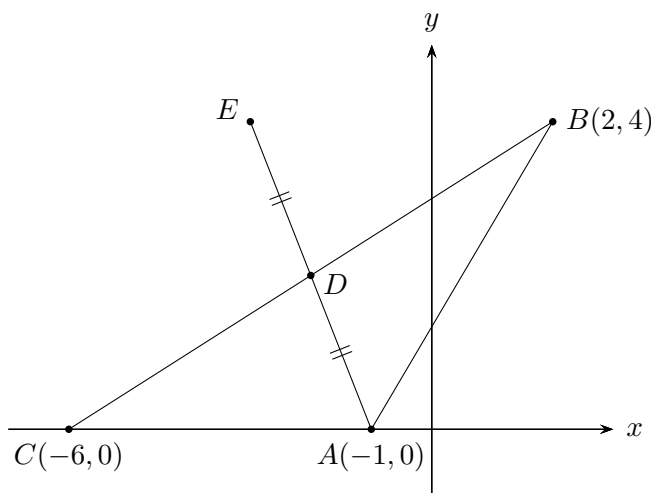
- (a) Differentiate with respect to x :
- i. $e^{\tan x}$. **2**
 - ii. $\frac{3x}{x^2 + 1}$. **2**
- (b) For the equation $3x^2 - 2x + 7 = 0$, evaluate:
- i. $\alpha + \beta$ **1**
 - ii. $\alpha\beta$ **1**
 - iii. $\alpha^2 + \beta^2$ **2**
- (c) A function is defined by $f(x) = x^3 - 3x^2 - 9x + 22$.
- i. Find the coordinates of the turning points of the graph $y = f(x)$ and determine their nature. **3**
 - ii. Find the coordinates of the point(s) of inflexion. **2**
 - iii. Hence sketch the graph of $y = f(x)$, showing the turning points, the point(s) of inflexion and the y intercept. **2**

Question 13 (15 Marks)

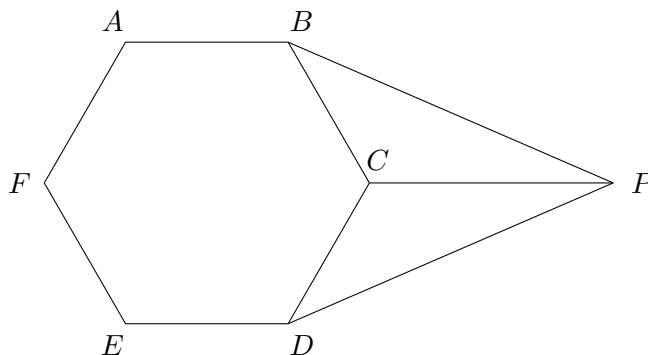
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Marks

- (a) In the diagram A , B , C and D are the points $(-1, 0)$, $(2, 4)$, $(-6, 0)$, $(-2, 2)$ respectively. D is also the midpoint of AE .



- i. Find the length of the interval AB . 1
 - ii. Find the equation of the circle with centre at B which passes through the point A . 1
 - iii. Find the size of $\angle CAB$ to the nearest degree. 2
 - iv. Find the midpoint of BC . 1
 - v. Show that the equation of the line BC is $x - 2y + 6 = 0$. 1
 - vi. Find the perpendicular distance of A from the line BC in simplest exact form. 2
 - vii. What type of quadrilateral is $ABEC$? Give reasons for your answer. 2
- (b) $ABCDEF$ Is a regular hexagon, and $CP \parallel AB$.



- i. Find the size of $\angle BCP$, giving reasons. 2
- ii. Prove that $\triangle BCP \equiv \triangle DCP$. 3

Question 14 (15 Marks)

Commence a NEW page.

Marks

- (a) For what value(s) of k does the equation **2**

$$x^2 + (k + 2)x + 4 = 0$$

have equal roots?

- (b) Evaluate the following integrals:

i. $\int \frac{1}{x\sqrt{x}} dx$ **2**

ii. $\int (\sin x + \cos x) dx$ **2**

- (c) Evaluate $\int_2^4 \frac{3x}{x^2 - 1} dx$, leaving your answer in the simplest exact form. **3**

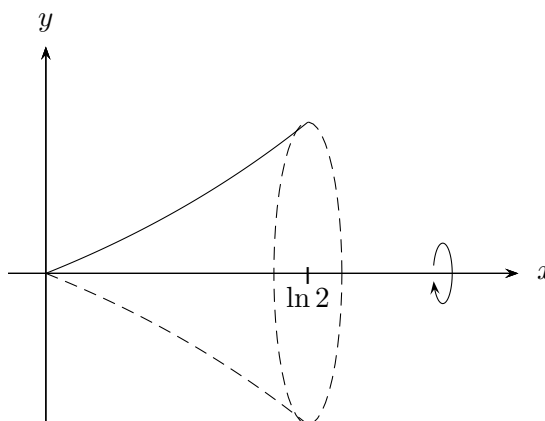
- (d) Consider the function $y = e^{x^2}$.

x	0	0.5	1.0	1.5	2.0
e^{x^2}	1.00		2.72		

- i. Copy the above table of values on to your page and supply the missing values (correct to 2 decimal places) **1**
- ii. Using Simpson's rule with five function values, find an estimate for the definite integral, correct to 2 decimal places: **2**

$$\int_0^2 e^{x^2} dx$$

- (e) The part of the curve $y = e^x - 1$ between $x = 0$ and $x = \ln 2$ is rotated about the x -axis. **3**



Find the exact volume (in simplest form) of the solid obtained.

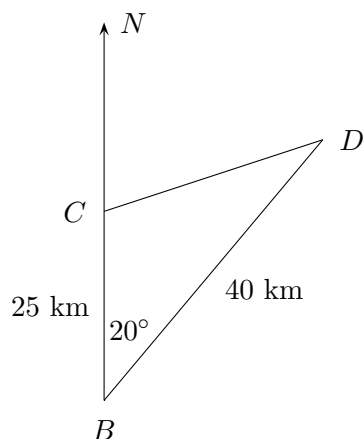
Question 15 (15 Marks)

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Marks

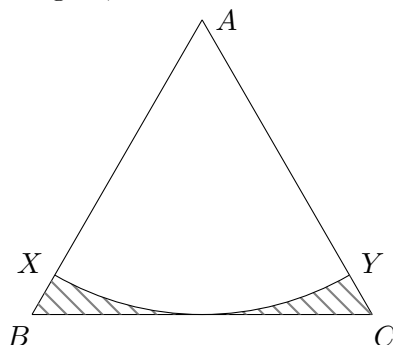
- (a) A town C is located 25 km due north of town B . Another town D is 40 km on a bearing of 020° from B . Towns B , C and D are connected by straight roads. **3**

Determine how much shorter it is for a man to travel from town D directly to town B , rather than through town C (give your answer correct to 1 decimal place).



NOT TO SCALE

- (b) In the diagram, $\triangle ABC$ is an equilateral triangle with sides of length 6 cm. An arc with centre A and BC as tangent, cuts AB and AC at X and Y respectively.



- i. Show that the radius of the arc is $3\sqrt{3}$ cm. **2**
 - ii. Find in exact form, the area of the shaded region. **3**
- (c)
- i. Differentiate $\log_e(\cos x)$ with respect to x , writing your answer in simplest form. **2**
 - ii. Sketch the curve $y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$. **2**
 - iii. Hence or otherwise, find the area bounded by the curve $y = \tan x$, the x axis and the line $x = \frac{\pi}{3}$, leaving your answers in simplest exact form. **3**

Question 16 (15 Marks)

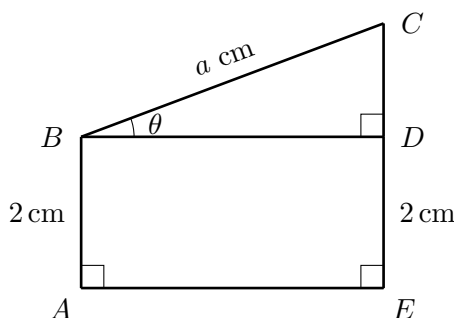
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Marks

- (a) In order to reduce the ‘bill shock’ that occurs whenever he receives his mobile phone bill, Mr Lam decides to reduce his mobile data usage by 15% of the previous day’s usage, starting from the first day of the billing cycle.

On the first day, he used 200 megabytes (MB) of mobile data.

- i. Find the amount of data (in MB) used on the fifth day, correct to 2 decimal places. **1**
 - ii. Find the total amount of data (in MB) used after 7 days, correct to 2 decimal places. **2**
 - iii. Show that a mobile phone plan which offers 1500MB of data per month would be sufficient for his usage. **2**
- (b) If $\tan \alpha = \frac{1}{2}$ and α is acute, find the *exact* value of $\sin \alpha$ and $\cos \alpha$. **2**
- (c) The figure shown represents a wire frame where $ABCE$ is a convex quadrilateral. D is a point on the line EC with $AB = ED = 2$ cm, and $BC = a$ cm, where $a > 0$.



Also, $\angle BAE = \angle CEA = \frac{\pi}{2}$, and $\angle CBD = \theta$, where $0 < \theta < \frac{\pi}{2}$.

- i. Find BD and CD in terms of a and θ . **2**
- ii. Find the length L , of the wire in the frame (which includes the length BD), in terms of a and θ . **1**
- iii. Find $\frac{dL}{d\theta}$, and hence show that $\frac{dL}{d\theta} = 0$ when $\tan \theta = \frac{1}{2}$. **2**
- iv. Given that $a = 3\sqrt{5}$, find the maximum length of wire in the frame. **3**

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

STUDENT NUMBER:

Class (please ✓)

☐ 12M2A – Mr Lowe

☐ 12M3A – Mr Lam

☐ 12M3B – Mr Berry

☐ 12M2B – Mrs Juhn

☐ 12M3C – Mr Lin

- | | | | | |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 5 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 6 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 7 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 8 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 9 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 10 – | ☐ A | ☐ B | ☐ C | ☐ D |

Suggested Solutions

(g) (3 marks)

Section I

1. (C) 2. (B) 3. (C) 4. (A) 5. (B)
6. (C) 7. (B) 8. (A) 9. (A) 10. (D)

Question 11 (Juhn)

(a) (2 marks)

$$\begin{aligned} 4x^2 - 36 &= 4(x^2 - 9) \\ &= 4(x - 3)(x + 3) \end{aligned}$$

(b) (2 marks)

$$\begin{aligned} |2x - 3| &< 13 \\ -13 &< 2x - 3 < 13 \\ +3 & \quad +3 \quad +3 \\ -10 &< 2x < 16 \\ -5 &< x < 8 \end{aligned}$$

(c) (2 marks)

$$2^{2x} - 9 \times 2^x + 8 = 0$$

Let $u = 2^x$,

$$\begin{aligned} u^2 - 9u + 8 &= 0 \\ (u - 8)(u - 1) &= 0 \\ \therefore u &= 8, 1 \\ \therefore 2^x &= 8, 1 \\ \therefore x &= 0, 3 \end{aligned}$$

(d) $(x - 2)^2 = 4y$

i. (1 mark) – $V(2, 0)$.

ii. (1 mark) – $y = -1$

(e) (2 marks)

$$D = \{x : x \neq 2, x \neq 3\}$$

(f) (2 marks)

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x - 2}{x^2 + x - 6} &= \lim_{x \rightarrow 2} \frac{\cancel{x - 2}}{\cancel{(x - 2)}(x + 3)} \\ &= \lim_{x \rightarrow 2} \frac{1}{x + 3} = \frac{1}{5} \end{aligned}$$

$$y = 2 \sin 2x$$

$$\begin{aligned} \frac{dy}{dx} &= 4 \cos 2x \Big|_{x=\frac{\pi}{8}} \\ &= 4 \cos \frac{\pi}{4} = \frac{4}{\sqrt{2}} \end{aligned}$$

Equation of tangent will be in the form
 $y = mx + b$:

$$y = \frac{4}{\sqrt{2}}x + b$$

When $x = \frac{\pi}{8}$, $y = \sqrt{2}$,

$$\begin{aligned} \sqrt{2} &= \frac{4}{\sqrt{2}} \times \frac{\pi}{8} + b \\ \therefore b &= \sqrt{2} - \frac{\pi}{2\sqrt{2}} \\ \therefore y &= \frac{4}{\sqrt{2}}x + \left(\sqrt{2} - \frac{\pi}{2\sqrt{2}} \right) \end{aligned}$$

Question 12 (Lam)

(a) i. (2 marks)

$$\frac{d}{dx} (e^{\tan x}) = (\sec^2 x) e^{\tan x}$$

ii. (2 marks)

$$\begin{aligned} y &= \frac{3x}{x^2 + 1} \\ u &= 3x \quad v = x^2 + 1 \\ u' &= 3 \quad v' = 2x \\ \frac{dy}{dx} &= \frac{vu' - uv'}{v^2} \\ &= \frac{3(x^2 + 1) - 2x(3x)}{(x^2 + 1)^2} \\ &= \frac{3x^2 + 3 - 6x^2}{(x^2 + 1)^2} = \frac{-3x^2 + 3}{(x^2 + 1)^2} \end{aligned}$$

(b) $3x^2 - 2x + 7 = 0$

i. (1 mark)

$$\alpha + \beta = -\frac{b}{a} = \frac{2}{3}$$

ii. (1 mark)

$$\alpha\beta = \frac{c}{a} = \frac{7}{3}$$

iii. (2 marks)

$$\begin{aligned}
 \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\
 &= \left(\frac{2}{3}\right)^2 - 2\left(\frac{7}{3}\right) \\
 &= \frac{4}{9} - \frac{14}{3} = -\frac{38}{9}
 \end{aligned}$$

(c) i. (3 marks)

$$\begin{aligned}
 y &= x^3 - 3x^2 - 5x + 22 \\
 y' &= 3x^2 - 6x - 9 \\
 &= 3(x^2 - 2x - 3) \\
 &= 3(x - 3)(x + 1)
 \end{aligned}$$

Stationary points occur when $y' = 0$:

$$\therefore x = -1, 3$$

When $x = -1$,

$$y = (-1)^3 - 3(-1)^2 - 9(-1) + 22 = 27$$

When $x = 3$,

$$y = 3^3 - 3(3^2) - 9(3) + 22 = -5$$

Finding the second derivative,

$$\begin{aligned}
 y' &= 3x^2 - 6x - 9 \\
 y'' &= 6x - 6
 \end{aligned}$$

When $x = -1$,

$$y' = 6(-1) - 6 < 0$$

$\therefore (-1, 27)$ is a local max. When $x = 3$,

$$y' = 6(3) - 6 > 0$$

$\therefore (3, -5)$ is a local min.

Alternatively, use table of variations:

x	-2	-1	0	3	4
$\frac{dy}{dx}$	$+$ 15	0	$-$ 9	0	$+$ 15
y		$(-1, 27)$		$(3, -5)$	

Hence $(-1, 27)$ is a local max, and $(3, -5)$ is a local min.

ii. (2 marks)

$$\begin{aligned}
 y' &= 3x^2 - 6x - 9 \\
 y'' &= 6x - 6
 \end{aligned}$$

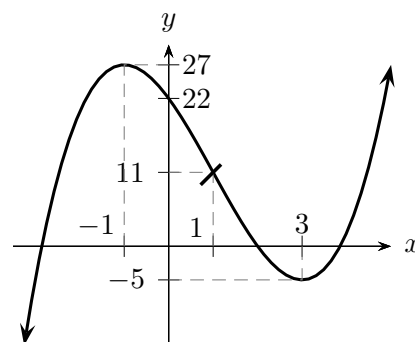
Pt of inflexion occurs when $y'' = 0$:

$$\begin{aligned}
 6x - 6 &= 0 \\
 x &= 1
 \end{aligned}$$

x	0	1	2
$\frac{d^2y}{dx^2}$	$-$ 6	0	$+$ 6
y		$\cap$	$\cup$

When $x = 1$, $y = 1 - 3 - 9 + 22 = 11$. Hence $(1, 11)$ is a point of inflexion as a change of sign of the 2nd derivative also occurs.

iii. (2 marks)

**Question 13** (Berry)

(a) i. (1 mark)

$$AB = \sqrt{(2+1)^2 + (4-0)^2} = 5$$

ii. (1 mark)

$$(x-2)^2 + (y-4)^2 = 25$$

iii. (2 marks)

$$m_{AB} = \tan \theta = \frac{4}{3}$$

$$\therefore \theta = 53.13^\circ \dots$$

$$\therefore \angle CAB = 180^\circ - 53.13^\circ \approx 127^\circ$$

iv. (1 mark)

$$M\left(\frac{2-6}{2}, \frac{4-0}{2}\right) = \left(-\frac{4}{2}, 2\right) = (-2, 2)$$

v. (1 mark)

$$\frac{y-0}{x+6} = \frac{4-0}{2+6} = \frac{1}{2}$$

$$\therefore y = \frac{1}{2}(x+6) = \frac{1}{2}x + 3$$

$$2y = x + 6$$

$$x - 2y + 6 = 0$$

vi. (2 marks)

$$(-1, 0) \rightarrow x - 2y + 6 = 0$$

$$d_{\perp} = \frac{|1(-1) - 2(0) + 6|}{\sqrt{1^2 + 2^2}}$$

$$= \frac{5}{\sqrt{5}} = \sqrt{5}$$

vii. (2 marks)

- As D is the midpoint of AE as well, $ABEC$ is a parallelogram as the diagonals bisect each other.
- But as $AC = 5$ as well as $AB = 5$, then $ABEC$ is a rhombus as adjacent sides are equal and also has the properties of a parallelogram.

Interior angle sum of polygon:
 $S = 180(n - 2)$

$$S = 180(6 - 2) = 720^\circ$$

$$\therefore \theta = \frac{720}{6} = 120^\circ$$

Hence $\angle ABC = 120^\circ$ and $\angle BCP = 120^\circ$ (alternate $\angle$ equal only if $AB \parallel CP$)

ii. (3 marks)

- Similarly, $\angle DCP = 120^\circ$.
- In $\triangle BCP$ and $\triangle DCP$:
- $BC = CD$ (sides of a regular hexagon)
 - $\angle BCP = \angle DCP$ (previously proven)
 - CP is common
- $\therefore \triangle BCP \equiv \triangle DCP$ (SAS)

Question 14 (Ziaziaris)

(a) (2 marks)

$$x^2 + (k+2)x + 4 = 0$$

$$\Delta = b^2 - 4ac$$

$$= (k+2)^2 - 4(1)(4)$$

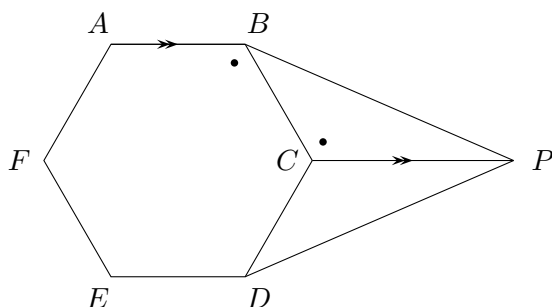
$$= (k+2)^2 - 16$$

$$= (k+2-4)(k+2+4)$$

$$= (k-2)(k+6)$$

Equal roots occur when $\Delta = 0$, i.e. when $k = 2$ or -6 .

(b) i. (2 marks)



(b) i. (2 marks)

$$\int \frac{1}{x\sqrt{x}} dx = \int x^{-\frac{3}{2}} dx$$

$$= -2x^{-\frac{1}{2}} + C$$

ii. (2 marks)

$$\int \sin x + \cos x dx = -\cos x + \sin x + C$$

(c) (3 marks)

$$\begin{aligned}
 \int_2^4 \frac{3x}{x^2-1} dx &= \frac{3}{2} \int_2^4 \frac{2x}{x^2-1} dx \\
 &= \frac{3}{2} [\log_e(x^2-1)]_2^4 \\
 &= \frac{3}{2} [\log_e(15) - \log_e(3)] \\
 &= \frac{3}{2} \log_e 5
 \end{aligned}$$

Question 15 (Lin)

(a) (3 marks)

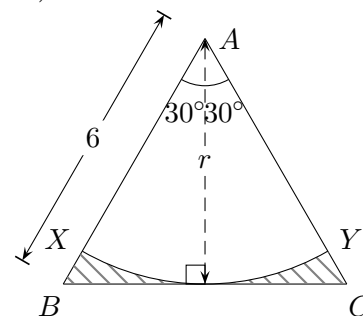
$$\begin{aligned}
 CD^2 &= 25^2 + 40^2 - 2(25)(40) \cos 20^\circ \\
 &= 345.61 \dots \\
 \therefore CD &\approx 18.59 \text{ km}
 \end{aligned}$$

$DC + CB = 43.59 \text{ km}$. Hence a difference of approximately 3.6 km.

(d) i. (1 mark)

x	0	0.5	1.0	1.5	2.0
e^{x^2}	1.00	1.28	2.72	9.49	54.6

(b) i. (2 marks)



ii. (2 marks)

$$\begin{aligned}
 \int_0^2 e^{x^2} dx &\approx \frac{h}{3} \left(y_0 + 4 \sum y_{\text{even}} + 2 \sum y_{\text{odd}} + y_\ell \right) \\
 &= \frac{\frac{1}{2}}{3} (1 + 4(1.28 + 9.49) + 2(2.72) + 54.6) \\
 &= 17.35
 \end{aligned}$$

$$\frac{r}{6} = \cos 30^\circ$$

$$\therefore r = 6 \cos 30^\circ = 6 \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$$

(NB. If student used exact values, their answer will be 19.61)

ii. (3 marks)

$$\begin{aligned}
 A_\Delta &= \frac{1}{2} ab \sin C \\
 &= \frac{1}{2} \times 6^2 \times \sin 60^\circ \\
 &= 18 \times \frac{\sqrt{3}}{2} = 9\sqrt{3}
 \end{aligned}$$

(e) (3 marks)

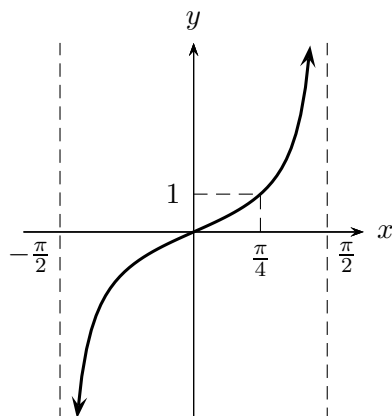
$$\begin{aligned}
 V &= \pi \int_0^{\log_e 2} (e^x - 1)^2 dx \\
 &= \pi \int_0^{\log_e 2} e^{2x} - 2e^x + 1 dx \\
 &= \pi \left[\frac{1}{2} e^{2x} - 2e^x + x \right]_0^{\log_e 2} \\
 &= \pi \left(\frac{1}{2} (e^{2 \log_e 2} - e^0) - 2(e^{\log_e 2} - e^0) + (\log_e 2 - 0) \right) \\
 &= \pi \left(\frac{1}{2} (4 - 1) - 2(2 - 1) + \log_e 2 \right) \\
 &= \pi \left(\frac{3}{2} - 2 + \log_e 2 \right) \\
 &= \pi \left(\log_e 2 - \frac{1}{2} \right)
 \end{aligned}$$

$$\begin{aligned}
 A_{\text{sect}} &= \frac{1}{2} r^2 \theta \\
 &= \frac{1}{2} (3\sqrt{3})^2 \times \frac{\pi}{3} \\
 &= \frac{1}{2} \times 27 \times \frac{\pi}{3} = \frac{9\pi}{2} \\
 A_{\text{shaded}} &= 9\sqrt{3} - \frac{9\pi}{2}
 \end{aligned}$$

(c) i. (2 marks)

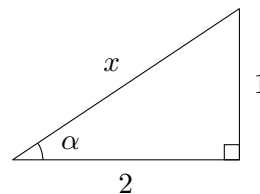
$$\frac{d}{dx} (\log_e \cos x) = \frac{-\sin x}{\cos x} = -\tan x$$

ii. (2 marks)



Maximum data usage would be 1 333.33 MB. Hence a 1 500 MB data plan would be sufficient.

(b) (2 marks)



iii. (3 marks)

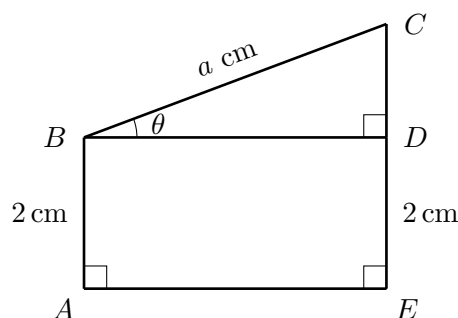
$$\begin{aligned}
 A &= \int_0^{\pi/3} \tan x \, dx \\
 &= - \int_0^{\pi/3} \frac{\sin x}{\cos x} \, dx \\
 &= - \left[\log_e \cos x \right]_0^{\pi/3} \\
 &= - \left(\log_e \cos \frac{\pi}{3} - \log_e \cos 0 \right) \\
 &= - \log_e \frac{1}{2} = \log_e 2
 \end{aligned}$$

$$x^2 = 1^2 + 2^2 = 5$$

$$\therefore x = \sqrt{5}$$

$$\therefore \sin \alpha = \frac{1}{\sqrt{5}} \quad \cos \alpha = \frac{2}{\sqrt{5}}$$

(c) i. (2 marks)

**Question 16** (Lowe)

(a) i. (1 mark)

$$\begin{aligned}
 a &= 200 \quad r = 0.85 \\
 T_5 &= ar^{n-1} \\
 &= 200(0.85)^4 \\
 &= 104.4 \text{ MB}
 \end{aligned}$$

ii. (2 marks)

$$\begin{aligned}
 S_7 &= \frac{a(r^n - 1)}{r - 1} \\
 &= \frac{200(0.85^7 - 1)}{0.85 - 1} \\
 &= 905.9 \text{ MB}
 \end{aligned}$$

iii. (2 marks)

- ✓ [1] for significant progress towards answer.
- ✓ [1] for final answer.

$$\begin{aligned}
 S &= \frac{a}{1 - r} \\
 &= \frac{200}{1 - 0.85} = 1\,333.33 \text{ MB}
 \end{aligned}$$

ii. (1 mark)

$$\begin{aligned}
 L &= 2BD + 2 + 2 + CD + a \\
 &= 2a \cos \theta + a \sin \theta + 4 + a
 \end{aligned}$$

iii. (2 marks)

- ✓ [1] for correct derivative.
- ✓ [1] for showing $\tan \theta = \frac{1}{2}$ when $\frac{dL}{d\theta} = 0$.

$$\frac{dL}{d\theta} = -2a \sin \theta + a \cos \theta$$

When $\frac{dL}{d\theta} = 0$,

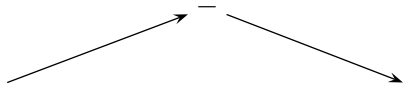
$$-2a \sin \theta + a \cos \theta = 0$$

$$\therefore \frac{2a \sin \theta}{\cancel{\div a \cos \theta}} = \frac{a \cos \theta}{\cancel{\div a \cos \theta}}$$

$$2 \frac{\sin \theta}{\cos \theta} = 1$$

$$\therefore \tan \theta = \frac{1}{2}$$

- iv. (3 marks) Maximum length of wire occurs when $\frac{dL}{d\theta} = 0$, i.e. $\tan \theta = \frac{1}{2}$ ($\theta \approx 0.46$)

θ	0	0.46	1
$\frac{dL}{d\theta}$	$\begin{smallmatrix} + \\ \cos 0 \end{smallmatrix}$	$\begin{smallmatrix} \\ 0 \end{smallmatrix}$	$\begin{smallmatrix} - \\ -2 \sin 1 + \cos 1 \end{smallmatrix}$
L			

NB. When $\theta = 1$,

$$\frac{dL}{d\theta} = a(-2 \sin 1 + \cos 1) \approx -1.14a$$

As $a > 0$, $\frac{dL}{d\theta} < 0$.

Hence $\tan \theta = \frac{1}{2}$ produces a local maximum.

$$\begin{aligned} \therefore L &= 2(3\sqrt{5}) \cos \theta + 3\sqrt{5} \sin \theta + 4 + 3\sqrt{5} \\ &= 6 \times \cancel{\sqrt{5}} \times \frac{2}{\cancel{\sqrt{5}}} + 3 \times \cancel{\sqrt{5}} \times \frac{1}{\cancel{\sqrt{5}}} + 4 + 3\sqrt{5} \\ &= 19 + 3\sqrt{5} \end{aligned}$$